

$$C_{rms} = \sqrt{\frac{3P}{\rho}}$$

Example 01:

Calculate the R.M.S. velocity of Hydrogen molecules at N.T.P. given: Density of Hydrogen at N.T.P. = $\rho = 8.957 \times 10^{-5} \text{ kg/m}^3$. Density of mercury = 13600 kg/m^3 , $g = 9.8 \text{ m/s}^2$. **1841**

Example 02:

Find the R.M.S. velocity of Nitrogen molecules at N.T.P. given that the density of Nitrogen at this temperature is 1.25 g/litre . **493.07**

Example 03:

Calculate the R.M.S. velocity of Oxygen molecules at a pressure of $1.013 \times 10^5 \text{ N/m}^2$ (N.T.P.) given the density of Oxygen is 1.44 kg/m^3 . **460**

Example 04:

Calculate the density of He at N.T.P. given that R.M.S. velocity of He molecules at N.T.P. is 1300 m/s . **6.19 kg/m^3**

Determine the pressure of oxygen at 0°C if the density of oxygen at NTP is 1.44 kg/m^3 and r.m.s speed of the molecule at NTP is 456.4 m/s . **10^5 N/m^2**

Example 06:

Two gases are at temperatures of 77°C and 27°C . What is the ratio of the R.M.S. velocities of the molecules of the two gases? **1.02:1**

Example 07:

At what temperature will the R.M.S. speed of the molecules of gas be three times its value at N.T.P.? **2457 K**

Example 08:

At what temperature will the R.M.S. speed of the molecules of gas be four times its value at N.T.P.? **4368 K**

Example 09:

Calculate the R.M.S. velocity of Oxygen molecules at 27°C . The density of Oxygen at N.T.P. is 1.44 kg/m^3 . **481.6**

Example 10:

Compute the R.M.S. velocity of Oxygen molecules at 127°C . Density of Oxygen at N.T.P. = **560**

Example 12:

Calculate the R.M.S. velocity of Oxygen molecules at 225°C . The density of oxygen at NTP is 1.42 kg/m^3 and $1 \text{ atmosphere} = 1.013 \times 10^5 \text{ N/m}^2$.

Example 12:

The density of a gas is 0.178 kg/m^3 at N.T.P. Find the R.M.S. velocity of gas molecules. By what factor will the velocity of molecules increase at 200°C ?

$$C \propto \sqrt{T}$$

$$\rho \propto \frac{1}{T}$$

$$\frac{\rho_1}{\rho_2} = \frac{T_2}{T_1}$$

$$273$$